

Ethics → principles & values → that help us differentiate betwⁿ right & wrong.

Moral Reasoning → ethical principle → think → decision making

- justice
- equality
- freedom
- health & safety

Social responsibility → ethical engineer will try to ensure 'no harm' to society, he will think about benefits of society.

Ethics in engineering

- socially responsible

- professionally

- code will be ethically built.

- profit gain in a proper way.

Typical ethical issues & examples:

- Safety - do not harm users

- acceptable risk - balance → innovation & risk.

Ex: Using self driving cars after safety testing

- compliance - follow laws, rules, regulations

- confidentiality - data privacy

- environment health - use ecofriendly goods

- data integrity

- conflict of interest - personal gain x professional judgement ✓

- Social impact

- Fairness - proper judgement

- Account for uncertainty - knowing the good & bad sides.

driverless cars → liability & responsibility

- - how safe are they
- infos
- hacking Δ, strong cybersecurity
- employment ↓

3 stages in the dev of Professional Identity

1) Independent operators → job sector → retirement

Professional life

↓

3 stages.

starting → job → guideline follow ~~also~~ develop

2) Team oriented → after 10/12 yrs work experience

code of ethics + earning respect of juniors

3) Self defining → after 20/22 yrs.

निर्णय ethical views + reflection of self.

mix of my - profession & ethics

Profession → Prev → moral / religious

17th cent → skills, capability first

judge / decide professionalism

1) Sociological account.

2) Social contract.

3) Morally permissible account.

- 1) Society gives special status / value
- 2) Professionals } money, trust, respect → good work for public.
Society }
- 3) Moral ethics

Characteristics of Profession

Public good → safety, health, welfare of public

↳ engineers should hold

- well being & quality of life
 - ACM → stated this
 - engineering ^{main aspect} → well-being of the public.
- code of ethics

* Five elements of well being.

Positive emotion

Enjoyment of activities

connection to something larger → sense of belonging to greater cause.

Accomplishment

Positive relationships.

* capabilities approach → Amartya Sen & Martha Nussbaum

→ well being & conditions for better.

* Well being & Engineering

→ CA + essential living condⁿs.

* Prohibited Actions from NSPE code

- not harm others
- clear rules through penalties or law enforcement (read from extra).

* Pre-requisite of well being

† Preventing harm to public

- potential harm identification
- harm → prevent

→ canon 1 → IEEE code of ethic

→ Engineers → take responsibility,
that affect public.

→ disclose risks

• case 82-5

- Public safety comes first

- Do not approve unsafe work

↳ risky work should not be approved.

→ board of ethical review

→ continuing to protest could lead to potential risk.

* Aspirational Ethics

- doing more than minimum requirement

- extra effort to some work for well being
of public or protect the environment.